

Store the bulk on the hard disks, but make sure your day-to-day essentials are on the cloud.

Security issues

A sudden realisation dawns when you notice your Facebook account containing personalised advertisements. Just how much information does one mouse click contain? This concern is exponential when it concerns storing sensitive data on the cloud. Is the so called off-site location manned by the third party server reliable and secure? Can you access the information on the cloud without other eyes tracking your every click? Can you restrict the recipients of the information on the cloud? These are few of the many concerns that plague users' minds as they reluctantly opt to store sensitive



data on the cloud rather than transfer it to a hard disk. To begin with, let's go through how the data on the web is secured. Most systems that do manage the data on the web, use a combination of techniques to ensure the security of information. Firstly, data is encrypted which is the process by which information sent is encoded in such a way that only a user with the encryption key can decode it. Most information

is encrypted by symmetric-key or public-key encryption. Symmetric-key encryption techniques have grown outdated, and public-key encryption techniques rule today. A popular implementation of this encryption technique is the Transport Layer Security (TLS) which is now accepted as an industry standard.

In the public-key encryption technique, the key used to decode information is based on a hash value. This hash value that is used, is computed from an input number using the hashing algorithm. The hashing algorithm provides a standard to convert an input number to the hash value. It is nearly impossible to extract the original input number from the hash value without knowing what the hashing algorithm does. Public keys use very sophisticated algorithms and large hash values for encrypting, including 40-bit or even 128-bit numbers. Trying to find a particular 128-bit number would be the equivalent of looking for the right grain of sand in the Sahara.

What encryption mainly does is that it assures a safe transit of data. To ensure that data sent is sent from a trustworthy source or the right one, authentication is used. The most common practice of authentication, though there exist several, that is implemented by cloud storage facilities is the